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WORK ON: DATA STORAGE AND RETRIEVAL SYSTEM IN HRIS

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DATA STORAGE AND RETRIEVAL SYSTEM IN HRIS

Introduction

A **Data Storage and Retrieval System** is one of the most important components of a Human Resource Information System (HRIS). It refers to the processes, technologies, and procedures used to store, organize, maintain, access, and retrieve employee information as required. Since human resources departments handle large volumes of employee-related data, an effective storage and retrieval system is essential to ensure that information remains accurate, secure, and readily available.

In the traditional paper-based environment, employee records were stored in filing cabinets and retrieved manually whenever needed. This process was time-consuming, expensive, and error-prone. With the introduction of HRIS, organizations can now store employee information electronically and retrieve it within seconds. This has significantly improved efficiency, accuracy, and decision-making in human resource management.

The data storage and retrieval system serves as the foundation of HRIS because all HR functions, such as recruitment, payroll administration, performance management, training, promotion, and retirement planning, depend on reliable employee information. If data is poorly stored or difficult to retrieve, HR managers may make incorrect decisions that can affect both employees and the organization.

Objectives of Data Storage and Retrieval in HRIS

The main objectives of a data storage and retrieval system include:

- Maintaining accurate and up-to-date employee records.
- Ensuring quick access to information when needed.
- Supporting managerial and strategic decision-making.
- Reducing paperwork and manual record keeping.
- Protecting employee information from unauthorized access.
- Supporting legal and regulatory compliance.
- Preserving historical records for future reference.
- Improving operational efficiency within the HR department.

Techniques of Data Retrieval in HRIS

Data retrieval techniques are the methods used to access and obtain information from the HR database. The effectiveness of an HRIS depends not only on how data is stored but also on how quickly and accurately it can be retrieved.

a) Search Technique

The search technique is the most common method of retrieving data. It involves locating employee information using a specific identifier or keyword. The HRIS searches the database and returns the matching records.

Common search criteria include:

- Employee ID number
- Employee name
- Department
- Job title
- Employee number
- National identification number

For example, if an HR officer wants to access the file of an employee named "John Smith," they can simply enter the employee's name or identification number into the system. The HRIS will immediately display the employee's record.

Advantages of the Search Technique

- Fast retrieval of information.
 - Easy for users to understand.
 - Reduces time spent searching manually.
 - Improves efficiency in HR operations.
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b) Query-Based Retrieval Technique

A query is a request made to the database to retrieve information that meets specific conditions. Unlike simple searching, queries can be used to analyze data and answer more complex questions.

For example, management may want to know:

- Which employees have worked for more than ten years?
- Which employees are due for promotion?

- How many employees are approaching retirement age?
- Which employees have attended training programs during the year?

The HRIS processes the query and produces the requested information.

Example

Retrieve employees from the Finance Department:

```
SELECT * FROM Employees
```

```
WHERE Department = 'Finance';
```

This technique is particularly useful for HR planning and decision-making because it allows managers to obtain specific information from large databases.

Advantages of Query-Based Retrieval

- Flexible and customizable.
- Supports strategic analysis.
- Provides accurate and detailed information.
- Facilitates evidence-based decision-making.

c) Report Generation Technique

Report generation involves producing organized summaries of employee information from the database. Reports are generated automatically by the HRIS and can be viewed on-screen or printed.

Common HR reports include:

- Payroll reports.
- Employee attendance reports.
- Leave management reports.
- Recruitment reports.
- Training reports.
- Performance appraisal reports.
- Employee turnover reports.

For example, at the end of each month, the HR department may generate an attendance report showing employees who were absent, late, or on leave.

Advantages of Report Generation

- Provides summarized information.
 - Saves time and effort.
 - Supports monitoring and evaluation.
 - Improves communication between HR and management.
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d) Dashboard Retrieval Technique

Modern HRIS systems use dashboards to display information visually through charts, graphs, and indicators. Dashboards allow managers to view important workforce information in real time without generating separate reports.

Examples of dashboard information include:

- Total number of employees.
- Employee turnover rates.
- Gender distribution.
- Training completion rates.
- Leave utilization rates.
- Recruitment progress.

A dashboard presents information in an easy-to-understand, easy-to-interpret format.

Advantages of Dashboard Retrieval

- Real-time information access.
 - Visual representation of data.
 - Faster decision-making.
 - Easy identification of trends and patterns.
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Types of Data Storage Systems in HRIS

Organizations use different types of storage systems depending on their size, resources, and operational requirements.

a) Centralized Data Storage System

A centralized storage system stores all employee information in a single database located at a central point within the organization.

In this system, all departments access the same database through a network. Changes made to employee records are immediately reflected across the organization.

Characteristics

- Single database.
- Centralized management.
- Standardized data structure.

Advantages

- Easy data management.
- Reduced duplication of records.
- Improved consistency of information.
- Better control over security policies.

Disadvantages

- System failure may affect all users.
- Higher risk if the central database is compromised.

b) Distributed (Decentralized) Data Storage System

In a distributed storage system, employee data is stored in multiple locations, such as branch offices or regional databases. These databases are connected through a network.

For example, a multinational company may maintain separate HR databases for each country while allowing data sharing between locations.

Advantages

- Faster local access to information.
- Reduced workload on a single server.
- Increased operational flexibility.

Disadvantages

- Difficult synchronization of records.
- Increased maintenance requirements.

- Risk of inconsistent information.
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c) Cloud-Based Data Storage System

Cloud storage involves storing employee data on remote servers accessible through the internet. The infrastructure is managed by cloud service providers.

Organizations increasingly prefer cloud-based HRIS because they reduce the need for expensive hardware and enable remote access.

Characteristics

- Internet-based storage.
- Accessible from multiple locations.
- Managed by external providers.

Advantages

- Accessibility from anywhere.
- Automatic software updates.
- Lower infrastructure costs.
- Easy scalability as the organization grows.

Disadvantages

- Dependence on internet connectivity.
 - Potential concerns about data privacy.
 - Risk of service interruptions.
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d) Hybrid Data Storage System

A hybrid system combines both on-premises storage and cloud storage. Some information is stored locally, while other information is stored in the cloud.

For example, sensitive payroll information may be stored internally, while training records are maintained in cloud-based applications.

Advantages

- Greater flexibility.
- Enhanced security.

- Improved disaster recovery.
- Better performance optimization.

Disadvantages

- More complex to manage.
 - Requires higher technical expertise.
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Methods of Data Storage in HRIS

Methods of data storage refer to the specific ways in which information is physically and logically stored within the HRIS.

a) Electronic Database Storage

Electronic databases are the most common method of storing HR information. A database organizes information into tables, records, and fields that can be easily searched and updated.

Examples of database systems include:

- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL

Employee records are stored systematically, making retrieval efficient and accurate.

Benefits

- High storage capacity.
 - Fast data retrieval.
 - Reduced redundancy.
 - Improved data integrity.
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b) File-Based Storage

File-based storage involves keeping HR information in electronic documents, such as:

- Microsoft Word files.

- PDF documents.
- Excel spreadsheets.
- Scanned employee records.

Although still used in some organizations, file-based systems are generally less efficient than database systems.

Benefits

- Easy to create and maintain.
- Low implementation costs.

Limitations

- Difficult searching.
 - Increased duplication.
 - Limited analytical capabilities.
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c) Data Warehouse Storage

A data warehouse is a large repository that stores historical HR data collected from multiple systems over a long period.

Unlike operational databases that support daily transactions, data warehouses are designed for analysis and strategic planning.

Applications

- Workforce forecasting.
- Employee turnover analysis.
- Succession planning.
- Compensation trend analysis.
- Performance trend evaluation.

Benefits

- Supports business intelligence.
 - Facilitates long-term planning.
 - Enables predictive analysis.
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d) Backup Storage Method

Backup storage involves creating duplicate copies of HR data to prevent data loss from hardware failures, cyberattacks, natural disasters, or human error.

Organizations typically maintain backup copies at separate locations.

Types of Backups

Full Backup

Copies all HR data.

Incremental Backup

Copies only data that has changed since the previous backup.

Differential Backup

Copies all changes made since the last full backup.

Importance

- Protects organizational information.
- Supports business continuity.
- Facilitates disaster recovery.

Importance of Data Storage and Retrieval Systems in HRIS

An effective storage and retrieval system contributes significantly to organizational success.

a) Faster Access to Information

HR professionals can retrieve employee information within seconds rather than spending hours searching through paper files. This improves service delivery and organizational responsiveness.

b) Improved Decision-Making

Managers rely on accurate information when making decisions about recruitment, promotions, transfers, compensation, and training. Quick access to reliable data enhances decision quality.

c) Increased Efficiency and Productivity

Automated storage and retrieval eliminate repetitive manual tasks, allowing HR personnel to focus on more strategic activities such as talent management and workforce planning.

d) Enhanced Data Security

Employee records contain sensitive information such as salaries, medical records, and personal details. HRIS protects this information through:

- Password protection.
- Encryption.
- Access controls.
- User authentication.
- Audit trails.

These measures reduce the risk of unauthorized access and data breaches.

e) Improved Accuracy and Reliability

Electronic systems reduce human error associated with manual recordkeeping. Validation mechanisms ensure that information entered into the system is accurate and complete.

f) Better Legal and Regulatory Compliance

Organizations are required to maintain employment records for legal purposes. HRIS helps ensure compliance with labor laws, tax regulations, pension requirements, and employment standards.

g) Cost Reduction

Digital storage reduces costs associated with:

- Paper.
- Printing.
- Physical storage facilities.
- Filing cabinets.

- Administrative labor.

Over time, organizations achieve significant cost savings.

h) Support for Strategic Human Resource Planning

Historical data stored in HRIS helps organizations analyze workforce trends and predict future staffing requirements. Managers can identify skill gaps, forecast retirements, and develop succession plans.

i) Improved Employee Services

Employees frequently request information regarding salaries, leave balances, benefits, and employment history. HRIS allows HR staff to respond quickly and accurately to such requests, improving employee satisfaction.

j) Preservation of Organizational Memory

Employee records, performance histories, and training data represent valuable organizational knowledge. Proper storage ensures that this information remains available for future reference and decision-making.

Conclusion

The **Data Storage and Retrieval System** is a fundamental component of Human Resource Information Systems because it ensures that employee information is stored securely, organized systematically, and retrieved efficiently whenever needed. Through various retrieval techniques such as searching, querying, reporting, and dashboard analysis, organizations can access accurate information for both operational and strategic purposes. Furthermore, different storage systems and methods—including centralized databases, cloud storage, data warehouses, and backup systems—enable organizations to manage growing volumes of HR data effectively. Ultimately, an efficient data storage and retrieval system improves decision-making, enhances security, increases productivity, reduces costs, and contributes significantly to the overall effectiveness of human resource management.